Assignment 5: Database Creation and Table Implementation

Database: MySQL | Project: Airline Reservation System

1. Objective

Create the MySQL database and implement the required tables using DDL, including their relationships.

2. Create database and tables

Run the following script in MySQL 8.0.16 or later. CHECK constraints are enforced in this version and later.

```
CREATE DATABASE airline_reservation_db;
USE airline_reservation_db;

CREATE TABLE Passengers (
    passenger_id INT PRIMARY KEY,
    full_name VARCHAR(100) NOT NULL,
    email VARCHAR(120) NOT NULL UNIQUE,
    phone VARCHAR(15) NOT NULL,
    passport_no VARCHAR(20) NOT NULL UNIQUE
);

CREATE TABLE Aircraft (
    aircraft_id INT PRIMARY KEY,
    model VARCHAR(50) NOT NULL,
    capacity INT NOT NULL,
    airline_name VARCHAR(100) NOT NULL
);

CREATE TABLE Flights (
    flight_id INT PRIMARY KEY,
    flight_number VARCHAR(20) NOT NULL UNIQUE,
    departure_airport VARCHAR(100) NOT NULL,
    arrival_airport VARCHAR(100) NOT NULL,
    departure_time DATETIME NOT NULL,
    arrival_time DATETIME NOT NULL,
    aircraft_id INT NOT NULL,
    FOREIGN KEY (aircraft_id) REFERENCES Aircraft(aircraft_id)
);

CREATE TABLE Bookings (
    booking_id INT PRIMARY KEY,
    booking_date DATE NOT NULL,
    seat_number VARCHAR(10) NOT NULL,
    travel_class VARCHAR(20) NOT NULL,
    passenger_id INT NOT NULL,
    flight_id INT NOT NULL,
    FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id),
    FOREIGN KEY (flight_id) REFERENCES Flights(flight_id)
);

CREATE TABLE Payments (
    payment_id INT PRIMARY KEY,
    payment_date DATE NOT NULL,
    amount DECIMAL(10,2) NOT NULL DEFAULT 0.00,
    payment_method VARCHAR(20) NOT NULL,
    payment_status VARCHAR(20) NOT NULL DEFAULT 'Pending',
    booking_id INT NOT NULL UNIQUE,
    FOREIGN KEY (booking_id) REFERENCES Bookings(booking_id)
);
```

3. Table relationships

Relationship	Type	Meaning
Aircraft → Flights	One-to-many	One aircraft can operate many flights
Flights → Bookings	One-to-many	A flight can have many bookings
Passengers → Bookings	One-to-many	A passenger can make many bookings
Bookings → Payments	One-to-one	A booking has one payment record
Passengers ↔ Flights	Many-to-many	Resolved through the Bookings table

4. Verify table creation

```
SHOW TABLES;
DESCRIBE Passengers;
DESCRIBE Aircraft;
DESCRIBE Flights;
DESCRIBE Bookings;
DESCRIBE Payments;
```

Expected tables: Aircraft, Bookings, Flights, Passengers, Payments. DESCRIBE displays each table's columns, data types, nullability and key information.

5. Conclusion

The schema contains five related tables. Primary and foreign keys establish identity and relationships, while constraints support reliable airline reservation and payment data.